Satyam Awasthi

Closed-Loop Brain-Computer Interfaces | Sensing, Decoding & Control

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Research Profile

My research interests center on reliable closed-loop brain-computer interfaces, from physiological signal acquisition and intent decoding to uncertainty-aware control, feedback, and adaptation. My current work spans EEG-based decoding and robust estimation and control under uncertainty.

Education

University of California, Santa Barbara — Santa Barbara, CA
M.S. in Computer Science — Sep 2021 – Mar 2023
GPA: 3.91/4.00. Four Eyes Laboratory; advisors: Tobias Höllerer and Michael Beyeler.

Indian Institute of Technology, Kharagpur — Kharagpur, India
B.Tech. in Electrical Engineering — Jul 2016 – Jul 2020
GPA: 8.98/10.0. Minor in Computer Science.

Current Research Projects

SSVEP-Based EEG Navigation

Remote
Research Project — 2026 – present
Guidance: Dr. Camille Gontier (Inria).

- Built four-channel BioAmp/MG24 acquisition, a custom-built EEG cap, optical timing tools, and a live CCA decoding pipeline.
- Evaluated live CCA decoding and offline decision-threshold replay, identifying target-discrimination differences and trade-offs between command detection and false activations.
- Developing an EEG interface for a C++/OpenGL maze; simultaneous four-target control and neural error feedback remain planned.

Sensory Feedback for Bidirectional Neural Interfaces

Remote
Research Collaboration — 2026 – present
Guidance: Prof. Mohsen Rakhshan (University of Central Florida).

- Investigating how peripheral stimulation waveforms relate to EEG responses and perceived sensation, toward more natural sensory feedback for bidirectional interfaces.
- Testing whether single-epoch EEG classification can distinguish five stimulation waveforms, while examining generalization across participants and potential stimulation and task-related confounds.

Robust Output-Feedback MPC

Remote
Research Project — Jul 2026 – present
Guidance: Dr. Johannes Köhler (Imperial College London).

- Developing coupled estimation and tracking-error bounds for state-dependent uncertainty using a reproduced ROHMPC certificate.
- Computed a 6.68% final-time observer-radius reduction in a synthetic bound-propagation experiment; closed-loop validation remains ongoing.

EEG-to-Kinematics Decoding

Remote
Research Project — 2026 – present
Guidance: Prof. Slawomir Nasuto (University of Reading).

- Built run-separated evaluation of wrist-position prediction from 32-channel EEG, using training-only scaling.
- Compared six regression baselines; CNN-BiLSTM achieved R^2 = 0.809/0.783/0.748 for wrist X/Y/Z on held-out runs from one participant.
- Cross-participant transfer and broader validation remain ongoing.

Earlier Research Experience

Four Eyes Laboratory, UC Santa Barbara Sep 2021 – Mar 2023

Graduate Researcher - EyeTTS

Advisor: Prof. Tobias Höllerer; co-advisor: Prof. Michael Beyeler.

- Developed the Magic Leap 1 eye-tracking study framework and Python calibration and analysis pipeline for mixed-reality locomotion.
- Studied gaze errors, eye-movement behavior, and temporal alignment under target and head/body motion.
- Contributed two first-author poster papers at IEEE ISMAR Adjunct 2023 and IEEE VR Workshops 2024.

IIT Kharagpur 2016 – 2018

Undergraduate Robotics Research - AGV Group & Technology Robotix Society

- Contributed perception, sensor fusion, and navigation components to the Eklavya autonomous ground vehicle platform.
- Worked on robot coordination and drone projects through Technology Robotix Society.

Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing

Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, Bivas Mitra

COMSNETS, 2022

10.1109/COMSNETS53615.2022.9668531

Workshop and Poster Papers

Eye Tracking Performance in Mobile Mixed Reality

Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, Tobias Höllerer

IEEE VR Workshops (VRW), 2024

10.1109/VRW62533.2024.00321

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Michael Beyeler, Tobias Höllerer

IEEE ISMAR Adjunct, 2023

10.1109/ISMAR-Adjunct60411.2023.00104

Selected Research Projects

SmartWatch for Wall Writing (WatchScribe) 2019 – 2020

IIT Kharagpur. Inertial-signal reconstruction of wall writing.

- Combined orientation correction and unsupervised stroke separation without a pretrained handwriting model; evaluated human readability and geometric reconstruction. Published at COMSNETS 2022.

JitterNot Spring 2022

UC Santa Barbara. Learning-assisted MPC for adaptive video streaming.

- Combined offline LSTM throughput prediction with constrained online resolution selection; evaluated against Jitsi Meet under emulated network conditions.

Wearable Human Activity Recognition Nov 2019

IIT Kharagpur. DeepConvLSTM classification of inertial signals.

- Built a six-channel, 100 Hz streaming pipeline with convolutional feature extraction and LSTM modeling of overlapping five-second windows.

Industry Experience

Intuit Inc Mountain View, CA
Software Engineer 2 Mar 2024 – Jan 2026

- Built dual-stage agent pipelines for intent routing, structured tool execution, and generative UI composition.

Yahoo Inc Mountain View, CA
Software Dev Engineer II (IC3) Apr 2023 – Nov 2023

- Developed production Android systems for Yahoo Mail, including data-driven inbox commerce and expense-management features.

Coupang Global LLC Mountain View, CA
Software Engineering Intern Jun 2022 – Sep 2022

- Developed user-facing search and product-discovery features for Coupang's Android commerce platform.

Samsung Research Institute Noida, India
Software Engineer (CL 2) Sep 2020 – Sep 2021

- Developed Android platform and device applications involving media processing, privacy, and isolated user-storage architectures.

Teaching

University of California, Santa Barbara Santa Barbara, CA
Graduate Teaching Assistant Sep 2021 – Mar 2023

Six quarters across introductory CS, mobile development, and senior capstone courses; mentored and evaluated 300+ students.

- Led discussion sections for up to 60 students, developed assessments, delivered guest lectures, and guided capstone teams.
- Courses: CMPSC 16, 24, 184, and 189A/B.

Skills

Programming: Python, C++, MATLAB, Java, Kotlin, Git

Machine learning & signal processing: PyTorch, NumPy/SciPy, OpenCV, CNN/LSTM models, Spectral analysis, Digital filtering

Control & estimation: Robust output-feedback MPC, Contraction-based methods, EKF, Sensor fusion, Motion planning

Robotics & interfaces: ROS/ROS2, Gazebo, RViz, Unity, OpenGL, BioAmp, Optical stimulus measurement

Academic Service

- Peer Reviewer, ACM UIST 2024 — reviewed submissions on interactive systems, spatial tracking, and interface hardware.